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Equilibrium, flow shear and stability measurements in the Z-pinch

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Abstract

The stabilizing effect of a sheared axial flow is investigated in the ZaP flow Z-pinch experiment at the University of Washington. Long-lived, hydrogen Z-pinch plasmas are generated that are 1 m long with an approximately 10 mm radius and exhibit gross stability for many Alfvén transit times. Large magnetic fluctuations occur during pinch assembly, after which the amplitude and frequency of the fluctuations diminish. This stable behaviour continues for an extended quiescent period. At the end of the quiescent period, fluctuation levels increase in magnitude and frequency. Axial flow profiles are determined by measuring the Doppler shift of plasma impurity lines using a 20-chord spectrometer. Experimental measurements show a sheared flow that is coincident with low magnetic fluctuations during the quiescent period. The experimental flow shear exceeds the theoretical threshold during the quiescent period, and the flow shear is lower than the theoretical threshold at other times. The observed plasma behaviour and correlation between the sheared flow and stability persists as the amount of injected neutral gas and experimental geometry are varied. Computer simulations using experimentally observed plasma profiles show a consistent sheared flow stabilization effect. Plasma pinch parameters are measured independently to demonstrate an equilibrium consistent with radial force balance.

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(Some figures in this article are in colour only in the electronic version)

1. Introduction

The Z-pinch provides a simple magnetic confinement configuration for plasma that has many advantages both as a possible fusion reactor and as a test bed to conduct basic plasma science research. The equilibrium is described by the radial force balance

$$\frac{B_\theta}{\mu_0 r} \frac{d(r B_\theta)}{dr} = -\frac{dp}{dr}. \quad (1)$$

The pure Z-pinch is simply connected and has unity average beta. The only magnetic field present is the B_θ field generated by the plasma current. All magnetic field lines are closed, even though the plasma has open ends. However, the pure Z-pinch is classically unstable to the $m = 0$ sausage and $m = 1$ kink modes.

Conventional techniques to provide stability for the Z-pinch have drawbacks. The sausage mode can be stabilized if the pressure gradient is limited [1]. Controlling the pressure profile is difficult and the technique does not stabilize the kink

mode. An axial magnetic field can be applied to provide stability. However, the plasma current and the pressure are limited by the strength of the axial magnetic field according to the Kruskal–Shafranov limit [2, 3]. Furthermore, an axial magnetic field opens all field lines and connects the electrodes to all regions of the plasma. A close-fitting, conducting wall can provide stability if the wall is located close to the plasma edge, $r_{\text{wall}}/a = 1.2$ [4], which is incompatible with a hot, fusion-grade plasma. The influence of an axial magnetic field and a conducting wall on the stability of a Z-pinch is also investigated in a classical work [5], which also reveals a stabilizing effect when the conducting wall approaches the plasma radius.

Flow shear can stabilize the MHD modes in a pure Z-pinch without the drawbacks of the conventional stabilization techniques. An axial flow does not alter the radial force balance that describes a Z-pinch equilibrium, see (1). Since the plasma geometry is simple, i.e. described by a one-dimensional force balance, the Z-pinch configuration is ideal for studying stability characteristics and isolating flow shear stabilization physics.

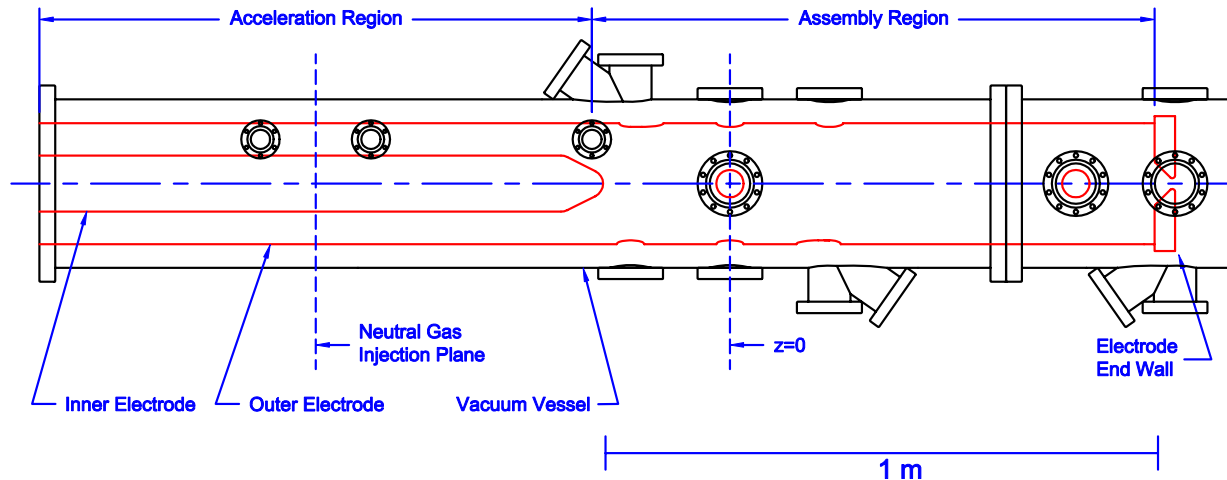


Figure 1. Side view machine drawing of the ZaP experiment identifying the relevant hardware features. The ports at $z = 0$ are used by the interferometers, the spectrometer for ion Doppler broadening measurements and the fast framing camera. These ports are also used for Thomson scattering measurements. The lower oblique port that views $z = 0$ is used for velocity profile measurements. The smaller side ports in the acceleration region are used to measure plasma density during acceleration. Positive z is to the right in the drawing. A 1 m scale is included for reference.

Linear stability analysis demonstrates a stabilizing effect of a sheared axial flow on the $m = 1$ kink instability in Z-pinchs when the shear exceeds a threshold value, $dV_z/dr > 0.1kV_A$ [4], assuming a uniform shear throughout the plasma. Other investigations have found that large values of the flow speed (Mach 2) are required for stability [6]. The investigations use different equilibrium velocity profiles. Reference [4] assumed a uniform shear throughout the plasma except at the axis and [6] assumed a shear that increases linearly from zero at the axis [7, 8].

Previous experiments have generated Z-pinch plasmas that exist for times longer than theoretically predicted by static plasma theory [9, 10]. These experiments have generated Z-pinch plasmas which inherently contain an axial plasma flow.

2. ZaP flow Z-pinch experiment

Experiments are conducted with the ZaP flow Z-pinch experiment at the University of Washington to investigate using sheared flows to provide stability in a pure Z-pinch. As described in [11], the ZaP flow Z-pinch experiment initiates a hydrogen plasma with a 1 m coaxial accelerator that has a 0.2 m diameter outer electrode and a 0.1 m diameter inner electrode. Neutral gas is injected through puff valves into the acceleration region. The plasma is accelerated to a large axial velocity by a Lorentz force, exits the accelerator and forms a Z-pinch plasma 1 m in length and approximately 10 mm in radius. Current in the accelerator continues to accelerate plasma into the Z-pinch assembly replacing plasma as it exits the Z-pinch. Inertia maintains the axial flow within the Z-pinch plasma column. A 10 mm radius hole is located along the axis in the electrode end wall to allow the axial plasma flow to escape from the assembly region and into a surge volume of the vacuum vessel. A machine drawing of the experiment is shown in figure 1, which identifies the relevant hardware features.

Experimental results reported here also include results from the recently modified ZaP experiment. The inner

electrode of the coaxial accelerator has been replaced with a larger diameter electrode, approximately 0.15 m. The new inner electrode also has added capacity and control for neutral gas injection.

3. Plasma fluctuations and velocity profile measurements

Diagnostics on the ZaP experiment are designed to measure the plasma flow profile and the stability of the pinch, as well as the plasma equilibrium parameters. Plasma stability is diagnosed with an azimuthal array of eight magnetic probes that measures the plasma's magnetic structure. The magnetic field probes of the azimuthal array are embedded in the inner surface of the outer electrode at the ' $z = 0$ ' location shown in figure 1. The eight probes are uniformly distributed azimuthally (45° apart). Data from these probes are Fourier analysed to determine the time-dependent evolution of the low-order azimuthal modes ($m = 1, 2, 3$), as shown in figure 2. These probes also provide a measure of the average magnetic field (B_0) and the corresponding axial current through the plasma column. The Fourier components of the magnetic field are normalized by the average magnetic field. The current sheet passes $z = 0$ at $22 \mu\text{s}$, where the normalized magnetic fluctuation data begin in figure 2. Before this time, $B_0 = 0$ and the normalized data have no meaning. Large magnetic fluctuations occur during pinch assembly, after which the amplitude and frequency of the magnetic fluctuations diminish. This stable behaviour continues for $35\text{--}45 \mu\text{s}$. The quiescent period is defined when the normalized $m = 1$ fluctuation levels are sustained below a threshold of 0.2, which corresponds to a 1 cm displacement of the plasma current from the geometric axis. A $37 \mu\text{s}$ quiescent period is seen in figure 2. Fluctuations during assembly generally reduce the highest m first, as shown in figure 2. Fluctuations are likely due to the dynamics of the formation process. At the end of the quiescent period, the fluctuation levels again change character, increase in magnitude and frequency and remain until the end of the plasma pulse.

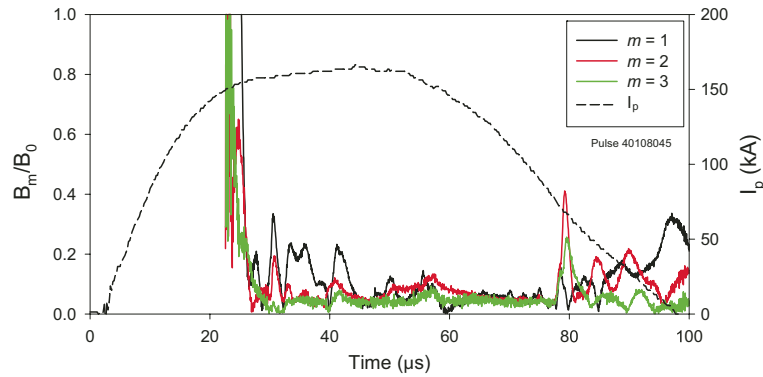


Figure 2. Time evolution of Fourier components of the normalized magnetic field fluctuation at $z = 0$ for the $m = 1, 2, 3$ modes for the original 0.1 m diameter inner electrode. The values are normalized to the average magnetic field value. A quiescent period is evident from 42 to 79 μs which defines the normalized time $\tau = 0$ to 1 for this pulse. The evolution of the total plasma current (dashed curve) is included for reference.

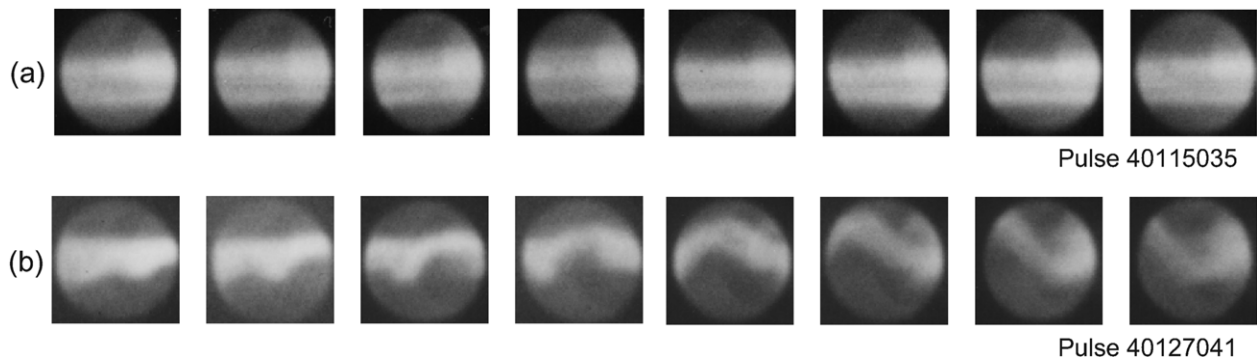


Figure 3. Fast framing camera images of visible light from the plasma viewed through a 5 cm hole at $z = 0$. Images are taken every 200 ns during a single plasma pulse. (a) Images obtained during 47.7–49.1 μs , the middle of the quiescent period. (b) Images obtained during 75.3–76.7 μs , near the end of the quiescent period.

The total plasma current is shown in figure 2 to identify the pulse length. The axial current in the plasma column during the quiescent period is approximately 50 kA. The remaining plasma current flows in the acceleration region and accelerates the plasma.

The magnetic fluctuation levels are correlated with fluctuation levels seen in visible photographs of the plasma. The plasma is viewed through a 5 cm hole in the outer electrode at $z = 0$, as shown in figure 1. The field of view is limited to a small volume in the middle of assembly region. Images are recorded using a fast framing camera (Imacon model 790) using a 200 ns framing module, which records an image every 200 ns for eight images during a single plasma pulse. The light is unfiltered, so the camera records broadband, visible light. The brightest line in the spectrum is H_α (656 nm), but other line radiation and bremsstrahlung also contribute to the recorded images. Figure 3(a) shows the evolution of the visible light images beginning at 47.7 μs with subsequent images to the right. The last image is obtained at 49.1 μs . The images are taken at a time during the quiescent period as defined by the magnetic fluctuation signals shown in figure 2. The z axis points to the left within each image, so the acceleration region is to the right of each image. The images show a stable plasma column throughout the images. Figure 3(b) presents images obtained during 75.3–76.7 μs , which is towards the end of the quiescent period as defined by the magnetic fluctuation signals

shown in figure 2. The images show the onset of an asymmetric 'kink-like' structure which grows throughout the images. The structure can be seen to travel across the 5 cm field of view in approximately 0.8 μs . It should be noted that at the end of the quiescent period the plasma typically disappears from the field of view. Large fluctuations of the plasma position during assembly also typically prevent images of the plasma assembly.

The plasma velocity profiles are determined by measuring the Doppler shift of impurity line radiation with a 0.5 m imaging spectrometer connected to an intensified charge-coupled device (ICCD) detector operated with a 100 ns gate. Typically, the C III impurity line at 229.7 nm is used. Light is collected along 20 parallel chords through the pinch plasma viewed at an oblique angle using the bottom, angled viewport that views the $z = 0$ plane, shown in figure 1. The 20 chords span 34 mm across the plasma. The intensity of the C III emission is typically peaked along the middle chords and decreases towards the outer chords, beyond the approximately 10 mm pinch radius [11]. This provides velocity data for the lower density plasma surrounding the pinch. The chord-integrated data are deconvolved using a shell/matrix method to give the radial profiles of the axial velocity [12]. Only one exposure of the ICCD camera is possible during a plasma pulse.

The trigger time of the ICCD camera is varied between pulses to determine the evolution of the velocity profiles. A

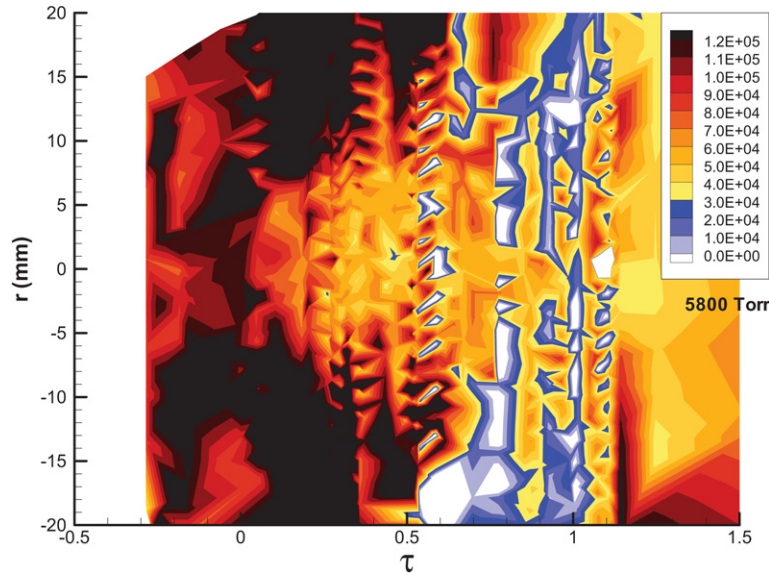


Figure 4. Contours of axial velocity (m s^{-1}) as a function of radius (for both sides of the axis) at $z = 0$ and of normalized time compiled for many pulses. A sheared plasma flow is evident during the quiescent period $0 < \tau < 1$. The gas line pressure is 5800 Torr.

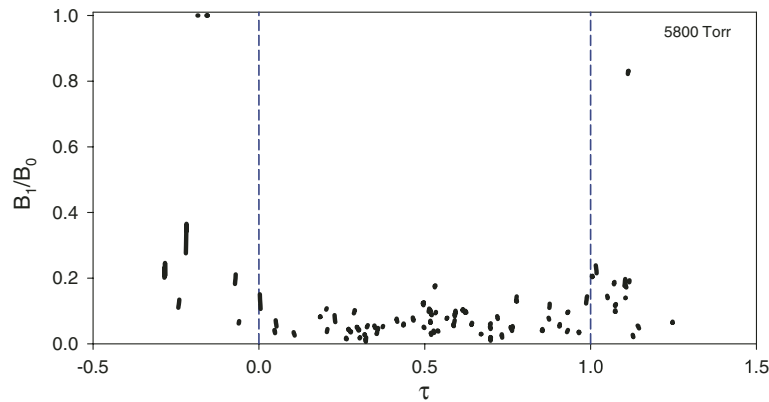


Figure 5. Magnetic mode data for $m = 1$ showing fluctuation levels collected during the same gate time as the spectroscopic data as shown in figure 4. The length of the data line segments for each time is indicative of the radial displacement of the plasma current during the ICCD gate time (100 ns).

normalized time, normalized to the quiescent period, is defined to account for small pulse-to-pulse variations and to allow detailed comparisons between the pulses. $\tau = 0$ is defined as the beginning and $\tau = 1$ is defined as the end of the quiescent period. The trigger times are also expressed using the normalized time τ . Figure 4 presents a contour plot of the deconvolved axial velocity as a function of radius (for both sides of the axis) and of normalized time compiled for many pulses. The flow profile shows a high, uniform axial flow before the quiescent period. The flow then develops a sheared flow structure with a high velocity ($\approx 10^5 \text{ m s}^{-1}$) at the plasma edge ($r \approx 10 \text{ mm}$) and a lower velocity ($\approx 5 \times 10^4 \text{ m s}^{-1}$) in the plasma core. The flow profile changes at approximately $\tau = 0.5$ when the edge velocity slows, while the core velocity remains approximately constant. At the end of the quiescent period, the flow profile shows a low, uniform axial flow. The corresponding magnetic mode data for $m = 1$ collected during the same 100 ns gate time as the spectroscopic data are shown in figure 5.

Density measurements are made with a holographic interferometer and with a two-chord He-Ne interferometer [13]. During the quiescent period a peaked electron density profile is measured with maximum values in the range of 10^{22} – 10^{23} m^{-3} . The interferometer also indicates a pinch radius of approximately 10 mm, beyond which the density decreases to less than half of the peak value. Using the experimentally measured density $n_e = 2 \times 10^{22} \text{ m}^{-3}$, pinch radius $a = 10 \text{ mm}$ and magnetic field at the pinch radius $B_a = 1 \text{ T}$, the Alfvén speed is calculated as $V_A = 1.5 \times 10^5 \text{ m s}^{-1}$. For a mode with a wavelength equal to the diameter, $ka = \pi$, the theoretical growth time for the kink mode in a static Z-pinch for these plasma parameters is $\tau_{\text{growth}} \approx (kV_A)^{-1} = 21 \text{ ns}$. The theoretical flow shear for these plasma parameters has a threshold value of $dV_z/dr > 0.1kV_A = 5 \times 10^6 \text{ s}^{-1}$.

The axial velocity is high and uniform during the assembly of the Z-pinch plasma ($\tau < 0$). The flow shear is $dV_z/dr = (0\text{--}4) \times 10^6 \text{ s}^{-1}$, as compared with the theoretical threshold $5 \times 10^6 \text{ s}^{-1}$ required for stability. At the start of the quiescent period $\tau = 0$, the velocity profile is high at the plasma edge and

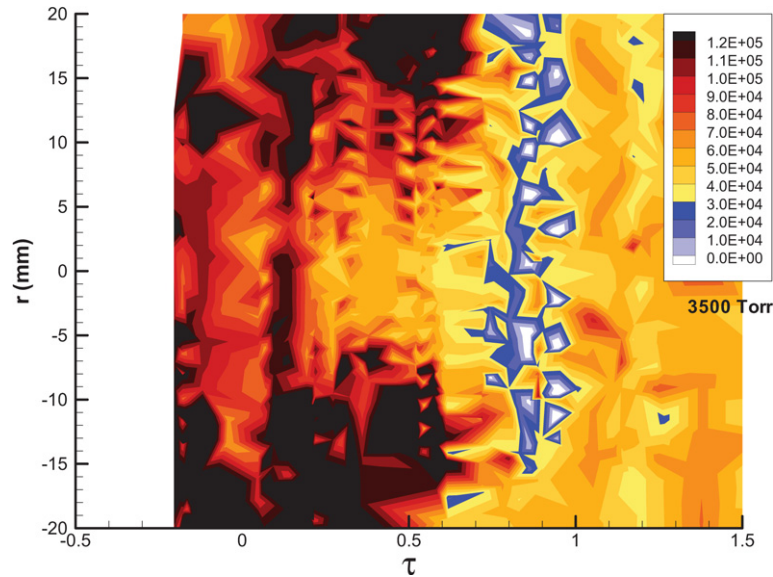


Figure 6. Contours of axial velocity (m s^{-1}) as a function of radius (for both sides of the axis) at $z = 0$ and of normalized time compiled for many pulses when the gas line pressure is reduced to 3500 Torr. A sheared plasma flow is evident during the quiescent period $0 < \tau < 1$.

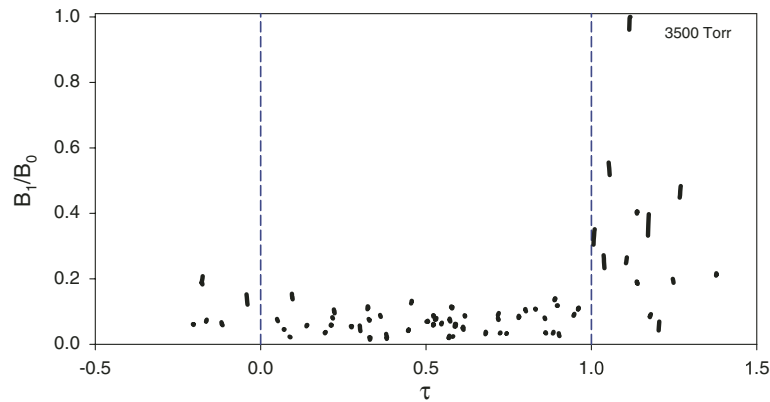


Figure 7. Magnetic mode data for $m = 1$ showing fluctuation levels collected during the same gate time as the spectroscopic data as shown in figure 6.

lower at the axis. The flow shear is $dV_z/dr = (7\text{--}12) \times 10^6 \text{ s}^{-1}$. The flow shear exceeds the theoretical threshold throughout the quiescent period, except for a brief time when the velocity at the edge slows so the velocity is higher along the axis and lower at the edge. At the end of the quiescent period $\tau = 1$, the velocity profile is low and uniform. The flow shear is $dV_z/dr = (0\text{--}6) \times 10^6 \text{ s}^{-1}$. The corresponding magnetic mode data shows low fluctuations during the sheared flow plasma state and high fluctuations before and after the quiescent period. Furthermore, the length of each of the data segments in figure 5 is indicative of the radial displacement of the plasma current during each ICCD gate time. The displacements are small during the quiescent period and large outside of the quiescent period. The general finding shows the experimental flow shear exceeds the theoretical threshold during the quiescent period, and the flow shear is lower than the theoretical threshold at times outside of the quiescent period [4, 11, 14, 15].

The amount of neutral gas injected is decreased and measurements are made of the flow profile evolution and plasma fluctuations, similar to the data presented in figures 4

and 5. The nominal gas line pressure is 5800 Torr. Experiments are conducted with gas line pressures at 4500, 3500 and 2500 Torr. All other run parameters are the same. The observed plasma behaviour and the correlation between the sheared plasma flow and the quiescent period are consistent for the variable amount of neutral gas. The results for 3500 Torr are shown in figures 6 and 7. A significant difference is that the quiescent period is seen to increase with the amount of injected neutral gas. During a plasma pulse, neutral gas continually ionizes, supplies plasma to the pinch and maintains the sheared flow Z-pinch state. This effect is measured with an axial array of 16 magnetic probes which are embedded in the surface of the outer electrode and span its length. The data show radial currents distributed throughout the acceleration region. The lower gas line pressure produces shorter quiescent periods, which suggests that the plasma source is being exhausted.

Measurements of the plasma velocity profile and magnetic fluctuations are repeated with the modified experimental geometry, where the 0.1 m diameter inner electrode of the coaxial accelerator has been replaced with an approximately

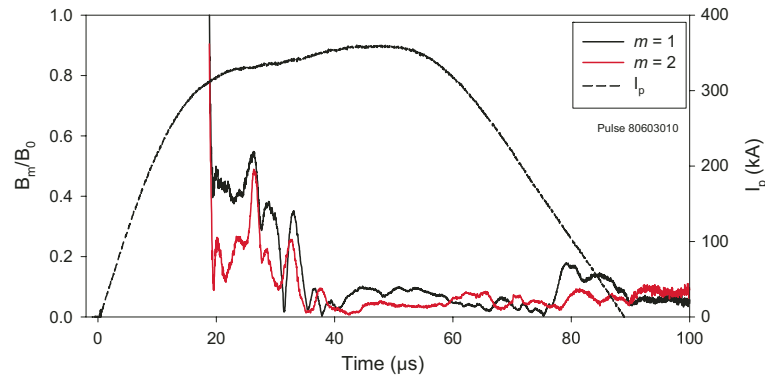


Figure 8. Time evolution of Fourier components of the normalized magnetic field fluctuation at $z = 0$ for the $m = 1, 2$ modes for the modified 0.15 m diameter inner electrode. The values are normalized to the average magnetic field value. A quiescent period is evident from 34 to 78 μs which defines the normalized time $\tau = 0$ to 1 for this pulse. The evolution of the total plasma current (dashed curve) is included for reference.

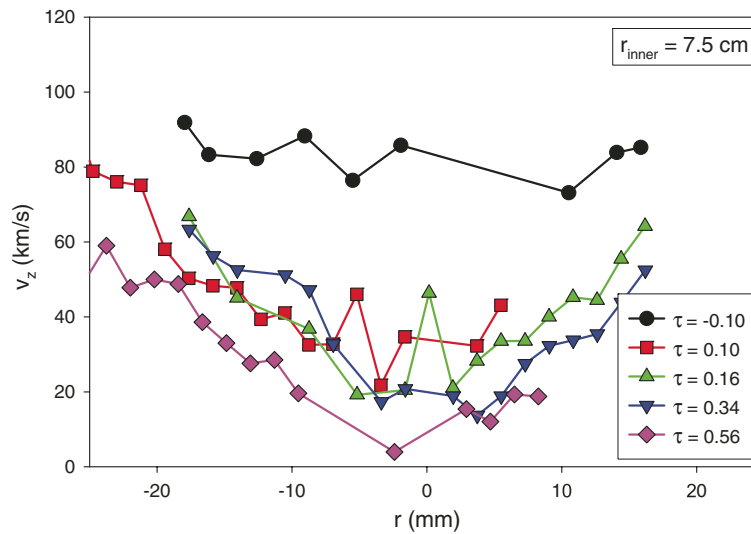


Figure 9. Axial velocity profiles (for both sides of the axis) at different values of normalized time for several pulses.

0.15 m diameter electrode. Magnetic fluctuation data are presented in figure 8, which shows a behaviour similar to that for the original experimental geometry. As in figure 2 for the original geometry, large magnetic fluctuations occur during pinch assembly, after which the amplitude and frequency of the magnetic fluctuations diminish. This stable behaviour continues for 34–78 μs . A 44 μs quiescent period is seen in figure 8. At the end of the quiescent period, the fluctuation levels again change character, increase in magnitude and frequency and remain until the end of the plasma pulse. The axial current in the plasma column during the quiescent period is approximately 75 kA. The stabilizing effect is evident in the modified experimental geometry, which has a 50% larger inner electrode, different neutral gas injection and a current through the plasma column of 75 kA compared with 50 kA for the original geometry. The total plasma current is higher in the modified geometry because of the lower inductance of the accelerator and the higher charge voltage, 9 kV compared with 5 kV.

The plasma velocity profiles in the modified experimental geometry are measured and deconvolved in the same manner as described previously. Figure 9 presents axial velocity profiles

at different values of normalized time for several pulses. As described earlier, the C III emission is finite in the lower density plasma beyond the pinch radius. The corresponding magnetic mode data for $m = 1$ collected during the same gate time as the spectroscopic data are shown in figure 10. The axial velocity is high and uniform during the assembly of the Z-pinch plasma ($\tau < 0$). The flow shear is $dV_z/dr = (0-2) \times 10^6 \text{ s}^{-1}$. Using experimentally measured density and magnetic field, the Alfvén speed is calculated. The theoretical growth time for the kink mode in a static Z-pinch is 25 ns for these plasma parameters. The theoretical flow shear required for stability for these plasma parameters has a threshold value of $4 \times 10^6 \text{ s}^{-1}$. During the quiescent period, $0 < \tau < 1$, the velocity profile is high at the plasma edge and lower at the axis. The flow shear is $dV_z/dr = (3-6) \times 10^6 \text{ s}^{-1}$. The corresponding magnetic mode data show low fluctuations during the sheared flow plasma state and high fluctuations before and after the quiescent period. The general finding is consistent with the original experimental geometry.

For both experimental geometries, the flow shear exceeds the theoretical uniform flow shear required for stability during the quiescent period, and the flow shear is below the threshold

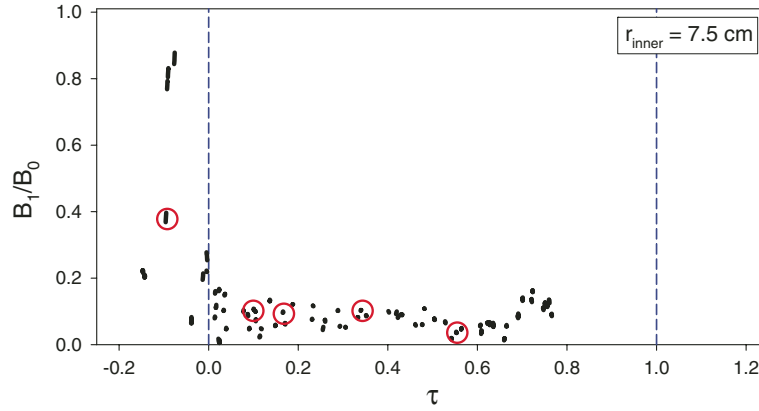


Figure 10. Magnetic mode data for $m = 1$ showing fluctuation levels collected during the same gate time as the spectroscopic data shown in figure 9 are circled. The length of the data line segments for each time is indicative of the radial displacement of the plasma current during the ICCD gate time.

when the plasma exhibits large fluctuations. However, the theoretical result is found for a uniform flow shear. The experimental measurements do not show a uniform flow shear. Furthermore, the theoretical threshold is sensitive to the plasma equilibrium, specifically the Alfvén speed, which is computed based on the magnetic field measured at the outer electrode and extrapolated with $1/r$ to the pinch radius. The effect of nonuniform flow shear is investigated computationally, and the validity of the assumed plasma equilibrium is investigated with additional experimental measurements. These are discussed in the next sections.

4. Z-pinch simulations with nonuniform flow shear

The flow velocity profiles experimentally measured throughout the quiescent period generally do not show a uniform flow shear across the pinch as assumed in the linear stability analysis. Instead, the velocity profiles typically show a flow shear that is localized at the edge of the pinch radius ($a = 10$ mm).

To test the effect of nonuniform flow shear on stability, nonlinear computer simulations of the $m = 0$ mode in a Z-pinch plasma are performed using Mach2 [16], a time-dependent, resistive MHD code. Three-dimensional simulations of additional modes ($m = 1, 2, 3$) would allow a more complete comparison with the experimental measurements and are the subject of a future study. An equilibrium is initialized with a parabolic pressure profile and an axially periodic density perturbation. A series of simulations are performed with varying plasma flow shear. Figure 11 shows Z-pinch simulation results (a) for a static plasma, (b) for a plasma with a uniform flow shear (with a peak $V_z = 0.2kaV_A$) and (c) for a plasma with a flow shear localized at the plasma edge (with a peak $V_z = 0.2kaV_A$).

The simulation results for a static plasma, figure 11(a), show the initial perturbation growing unstable until plasma confinement is lost. The static plasma simulation represents the reference case to compare with the simulations with finite flow shear. The simulation results shown in figure 11(b) are for a plasma with a uniform flow shear, with $V_z = 0$ at the axis and $V_z = 0.2kaV_A$ at the pinch radius with a linear variation with radius for $0 < r \leq a$ and a uniform flow beyond the pinch

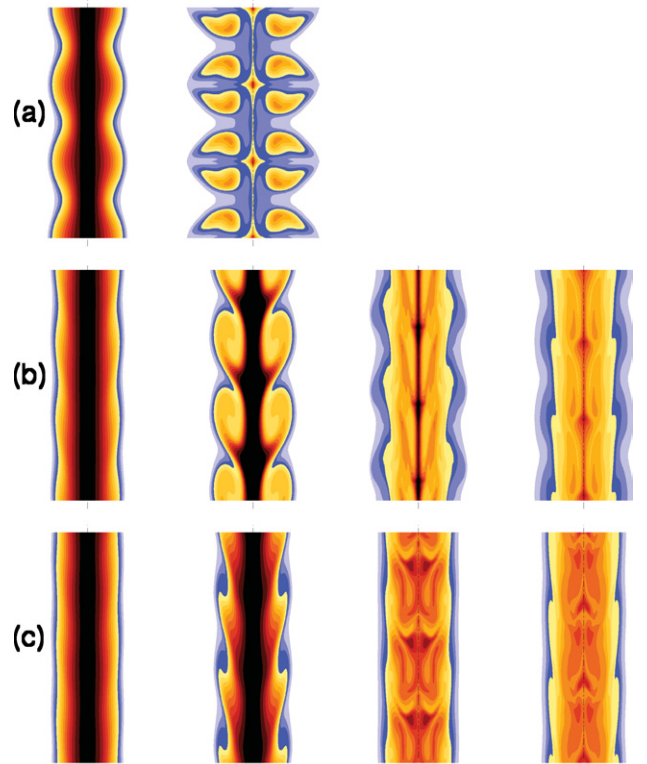


Figure 11. Nonlinear simulation results showing pressure contours in a Z-pinch. The results for each simulation time are aligned vertically. (a) No equilibrium axial flow is initialized. The plasma quickly develops an $m = 0$ mode and loses confinement. (b) An equilibrium axial flow with a uniform shear through the pinch is initialized. The $m = 0$ mode is significantly less developed than the static plasma case. (c) An equilibrium axial flow with a shear increasing towards the plasma edge is initialized. The stabilizing effect is again evident.

radius, $r > a$. The uniform flow shear produces a substantially reduced $m = 0$ instability as compared with the static plasma.

Figure 11(c) shows simulation results for a plasma with a flow shear that varies as r^5 , with $V_z = 0$ at the axis and $V_z = 0.2kaV_A$ at the pinch radius with a r^6 variation for $0 < r \leq a$ and a uniform flow beyond the pinch radius, $r > a$.

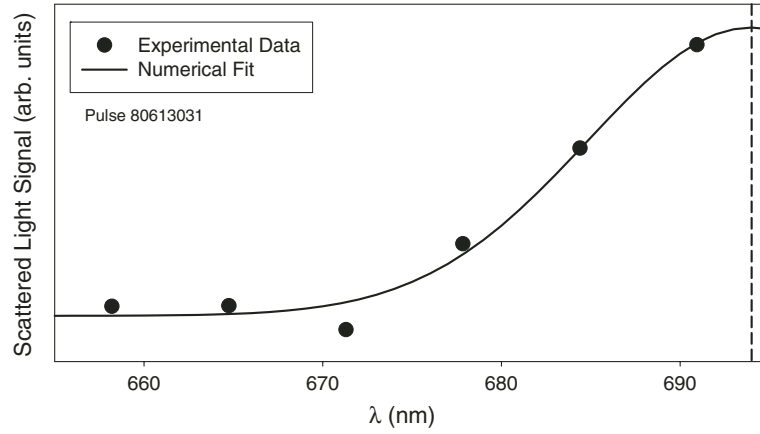


Figure 12. Thomson scattered spectrum measured by five wavelength channels. The data are fitted with a Gaussian line profile corresponding to a 64 eV electron temperature.

The stabilizing effect is still evident. The stabilizing results for a nonuniform shear are consistent with other research that indicates the local shear is important for stability [17]. Specifically, the shear must be located at the outer boundary of the plasma, as is the case for the simulation shown in figure 11(c).

5. Experimental equilibrium measurements

The theoretical threshold for the sheared flow stabilization is computed based on measurements of the plasma equilibrium using plasma density from interferometry, pinch radius from interferometry, and magnetic field extrapolated from magnetic probes in the outer electrode. The plasma electron and ion temperatures (T_e and T_i) are measured independently to check for agreement with the plasma parameters by using equilibrium force balance, (1). The equilibrium force balance can be expressed as

$$\frac{B_\theta}{\mu_0 r} \frac{d(r B_\theta)}{dr} = -\frac{d}{dr} [nk(T_e + T_i)] \quad (2)$$

for a charge-neutral ($n = n_e = n_i$) hydrogen plasma, where k is Boltzmann's constant.

The electron temperature is measured with a single-point Thomson scattering system. The diagnostic uses a 10 J Korad ruby laser which crosses the pinch plasma at $z = 0$ along a diameter using the horizontal viewports shown in figure 1. Scattered light is collected by fibres on the top viewport. The light collection system views a cylindrical scattering volume 2.5 mm long and 3 mm diameter at the machine axis along the focused laser beam waist. The scattered light is directed to a custom-built Hibshman Thomson scattering spectrometer which uses line filters to remove the parasitic ruby light (694 nm) and the H_α emission. The output field of the spectrometer is recorded with an array of eight photomultiplier tubes (PMTs) distributed across the scattered spectrum. The laser pulse and scatter pulse waveforms are recorded, as well as the background light before and after the scatter pulse. A representative scattered spectrum is shown in figure 12. The spectrum is fitted with a Gaussian line profile corresponding to a 64 eV electron temperature.

The ion temperature is determined by measuring the Doppler broadening of impurity lines caused by thermal motion. The same ICCD spectrometer that is used to make velocity profile measurements is also used to measure the ion temperature by viewing the plasma through a radial viewport. For these data, the bottom viewport at $z = 0$ in figure 1 is used to measure the broadened spectrum of the C III impurity line at 229.7 nm along a chord through the plasma diameter. The spectrum shown in figure 13 shows the experimentally measured spectrum and a numerical fit of a Gaussian line profile corresponding to a 71 eV ion temperature. The spectrum is recorded early during the quiescent period during a 100 ns gate.

During the same gate time used for the ICCD, data from a four-chord interferometer through the pinch plasma are analysed. The four-chord interferometer is an expanded version of the two-chord interferometer used on the original geometry [13]. Assuming a uniform density inside the pinch surrounded by a uniform background density gives a pinch density of $n_e = 1.5 \times 10^{22} \text{ m}^{-3}$ and a background, surrounding density of $n_e = 2 \times 10^{21} \text{ m}^{-3}$ with a pinch radius $a = 13 \text{ mm}$. During this time, the magnetic field at the outer electrode is measured at 0.13 T. If no current flows outside of the pinch, the magnetic field at the pinch radius is $B_a = 1 \text{ T}$. For a uniform pressure inside the pinch, the equilibrium force balance gives a total temperature $T_e + T_i = 160 \text{ eV}$. This value can be compared with the independently measured electron and ion temperatures $T_e + T_i = 135 \text{ eV}$. The comparison suggests that the equilibrium values used to compute the theoretical shear threshold are valid.

6. Conclusions

The ZaP project is producing Z-pinch plasmas that exhibit gross stability during an extended quiescent period, consistent with flow stabilization theory. The quiescent period is coincident with a sheared plasma flow. The magnitude of the sheared plasma flow is consistent with flow stabilization theory, even for different amounts of injected neutral gas. Nevertheless, the stability results presented should be interpreted carefully. A stable plasma is observed coincident with a sheared flow; however, causality has not been

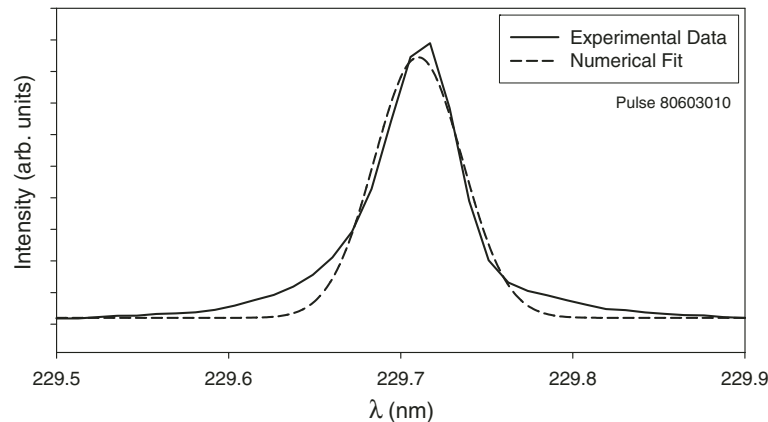


Figure 13. Doppler broadened C III impurity line spectrum at 229.7 nm. A numerical fit of a Gaussian line profile corresponding to a 71 eV ion temperature is also shown.

established. Furthermore, the effect of large Larmor radii can approximately halve the growth rate of the $m = 0$ mode for some equilibria [18]. The ratio of the average ion Larmor radius to the pinch radius is 0.1–0.2.

Neutral gas supply appears to limit the plasma lifetime and suggests a technique to produce long-lived plasmas. The behaviour remains consistent for the different amounts of injected neutral gas and as the diameter of the inner electrode of the coaxial accelerator is increased by 50%. Nonlinear computer simulations demonstrate stabilization with flow profiles that have a shear localized at the plasma edge, similar to experimental measurements. Recent measurements from multiple diagnostics demonstrate reasonable agreement among the experimental measurements with equilibrium force balance.

A flow-stabilized Z-pinch has many important implications for a simple reactor design with average unity β and a simply connected geometry. In addition, the stabilizing effect of sheared flows has important applications to other magnetic confinement concepts. Understanding the physical effects of sheared flows on stability is important to advancing our general understanding of plasma physics and specifically magnetic fusion science, beyond the flow Z-pinch concept alone.

Acknowledgments

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